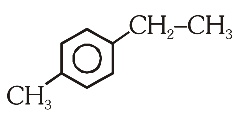
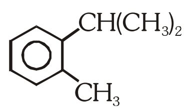
AROMATIC HYDROCARBANS

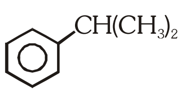
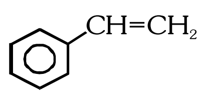
1. Which of the following is a polynuclear hydrocarbon
(1) Cumene (2) Napthalene
(3) Mesitylene (4) Toluene

2. An aromatic compound contains . . . . . . π electrons.
(1) 4n (2) 4n + 1
(3) 4n + 2 (4) 4n + 3

3. Number of π electrons present in Anthracene
(1) 6 (2) 10 (3) 14 (4) 18

4. Which of the following structure is called cumene

(1)  (2) 

(3)  (4) 

5. Cyclooctatetraene is not aromatic because
(1) It has four carobn-carbon double bonds
(2) It has not got (4n + 2) π electrons (where n = 1, 2, 3, ... etc) and is non planar
(3) It is planar in structure
(4) It decolourises potassium permagnate solution

6. $A \xrightarrow[\text{Cu tube}]{\text{Red hot Fe or}} \text{Benzene}$
Identify A :-
(1) Ethyne (2) 2-Butyne
(3) Propyne (4) Methanal

7. Which of the following groups of names refer to the same compound
(1) Lindane, 1,2,3,4,5,6-Hexachlorocyclohexane, B.H.C.
(2) Hexachlorobenzene, Benzyl chloride, Benzotrichloride
(3) Lindane, Benzylidene chloride, D.D.T
(4) Gammexane, B.H.C, D.D.T

8. Benzene is obtained by all the following reactions except
(1) Decarboxylation of sodiumbenzoate
(2) Deoxygenation of phenol
(3) Reduction of Benzene diazonium chloride with ethanol
(4) Catalytic hydrogenation of acetylene

9. Oxidation of which of the following arenes does not give benzoic acid
(1) Ethylbenzene (2) Cumene
(3) Mesitylene (4) Sec. Butylbenzene

10. Acetylation of Benzene takes place by –
(1) $Cl_2 + h\nu$
(2) $CH_3COCl + AlCl_3$
(3) $CH_3Cl + AlCl_3$
(4) $C_6H_5COCl + AlCl_3$

11. Benzene + $CH_3Cl \xrightarrow{A}$ Toluene, (A) is –
(1) Ni (2) $AlCl_3$ (3) Pd (4) pt

12. Benzene on treatment with n-Propyl chloride in presence of Anhydrous $AlCl_3$ mainly gives –
(1) Toluene (2) n-Propylbenzene
(3) Ethylbenzene (4) Cumene

13. When benzene is heated with acetic anhydride in presence of anhydrous $AlCl_3$, the product formed is –
(1) Benzoic acid
(2) Acetophenone
(3) Ethylphenyl ketone
(4) Benzophenone

14. Photochlorination of Benzene finally gives –
(1) Benzene hexachloride
(2) Chlorobenzene
(3) Benzyl chloride
(4) Benzylidene chloride

15. Which of the following reaction is called Wurtz-Fittig reaction ($\phi = C_6H_5$)
(1) $\phi\text{–X} + \text{R–X} \xrightarrow{AlCl_3} \phi\text{–R} + X_2$
(2) $\phi\text{–H} + \text{R–X} \xrightarrow{AlCl_3} \phi\text{–R} + HX$
(3) $\phi\text{–}CH_3 + SO_2Cl_2 \xrightarrow{\text{Peroxide}} \phi\text{–}CH_2Cl + SO_2 + Cl_2$
(4) $\phi\text{–X} + \text{R–X} \xrightarrow[\text{dry ether}]{\text{Na}} \phi\text{–R} + 2NaX$

16. Ozonolysis of Benzene yields –
(1) Glyoxal (2) Glycerol
(3) Glycol (4) Glycine

17. Which of the following acid on decarboxylation does not form an aromatic hydrocarbon

(1) Phenylacetic acid

(2) Salycylic acid

(3) Ortho toluic acid

(4) Benzoic acid

18. Which of the following is not correctly matched –

(1) Deoxygenation of phenol – Benzene

(2) Decarboxylation of p-toluic acid – Toluene

(3) Oxidation of Toluene – Benzoic acid

(4) Oxidation of Ethylbenzene – Phenol

19. The number of Benzylic hydrogen in ethylbenzene is –

(1) 3 (2) 5 (3) 2 (4) 7